Reciprocal Functions: Inequalities

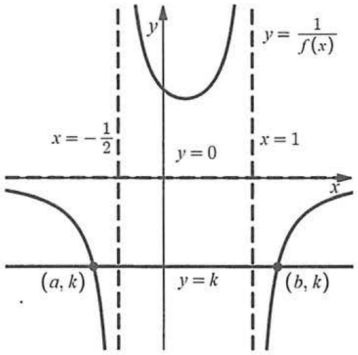
Questions 1–7

Question 1. By first sketching a graph, find the values when $\frac{1}{x+2} \leq 0$, giving your answer in set notation.

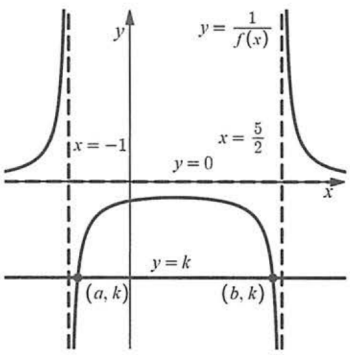
Question 2. Each of the following graphs show curves $y = \frac{1}{\mathrm{f}(x)}$ and the line $y = k$, for a constant k which intersect at the given points in terms of constants a and b

In each case, determine the values for which $\frac{1}{\mathrm{f}(x)} \leq k$, giving your answer in interval notation in terms of a and b

a)



b)



c)

Question 3. Solve each of the following equalities, giving your answer in interval notation:

a) $\frac{1}{x^2-4} \leq 0$

b) $\frac{1}{6+5x-x^2} < 0$

Question 4. Solve each of the following equalities, giving your answer in set notation:

a) $\frac{1}{x^2-9} > \frac{1}{6}$

b) $\frac{1}{x(x-1)(x+4)} \leq 0$

Question 5.

a) Show that $(4x-1)$ is a factor of

$$\mathrm{f}(x) = 12x^3 + x^2 - 9x + 2$$

b) Sketch $y = \frac{1}{12x^3+x^2-9x+2}$

c) Hence solve $\frac{1}{12x^3+x^2-9x+2} < 0$ giving your answer in set notation.

Question 6.

a) Show that $(3x-1)$ is a factor of

$$\mathrm{f}(x) = 6x^3 - 5x^2 - 2x + 1$$

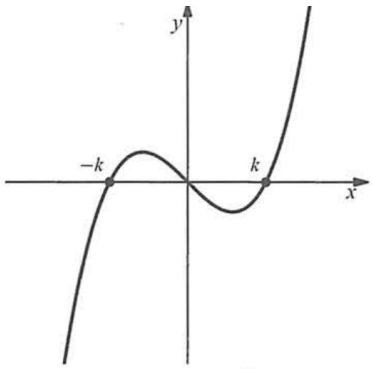
b) On the same set of axes sketch the $y = \frac{1}{6x^3-5x^2-2x+1}$ and $y = 1$
You do not need to label the points of intersection.

c) Hence solve $\frac{1}{6x^3-5x^2-2x+1} < 1$ giving your answer in interval notation.

Question 7. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = x(x-k)(x+k)$$

where k is a positive integer is shown in the diagram below.



a) Sketch the graph of $y = \frac{1}{\mathrm{f}(x)}$

b) Hence, solve $\frac{1}{x(x-k)(x+k)} > 0$
Give your answer in set notation.